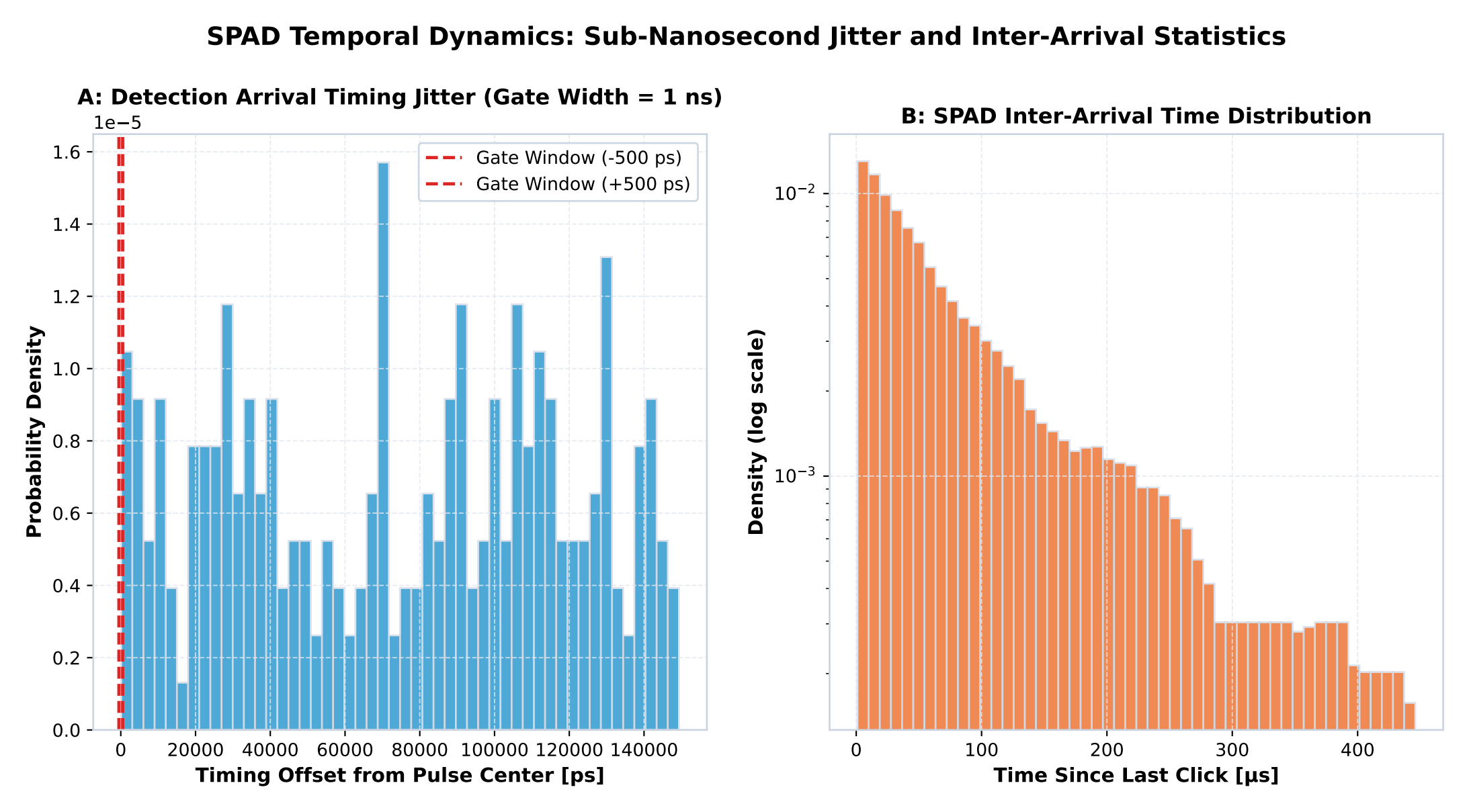
SPAD Temporal Dynamics: Sub-Nanosecond Jitter and Inter-Arrival Statistics
A: Detection Arrival Timing Jitter (Gate Width = 1 ns)
1e−5
Gate Window (-500 ps)
Gate Window (+500 ps)
Probability Density
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
0
20000
40000
60000
80000
100000
120000
140000
Timing Offset from Pulse Center [ps]
B: SPAD Inter-Arrival Time Distribution
Density (log scale)
$10^{-2}$
$10^{-3}$
0
100
200
300
400
Time Since Last Click [µs]